How Often Should A Liquidity Provider Recenter? A Block-time Replay Study

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Abstract

Decentralized finance (DeFi) has turned blockchains into open financial infrastructure: anyone can trade, lend, or provide liquidity through smart contracts without a central intermediary. Decentralized exchanges in this ecosystem typically use automated market makers (AMMs) rather than limit-order books—traders swap against an on-chain pool, and the pool's reserves set the price. Liquidity providers (LPs) supply the pool's assets and earn a share of trading fees in return.

Uniswap v3, the dominant concentrated-liquidity AMM, lets each LP deploy capital only inside a chosen price interval. While the market price stays in that interval, the position earns fees; once the price drifts out, fees stop. The LP must therefore decide how often to *recenter*—move the interval so that the current price sits at its center again—but on Ethereum this is a systems problem, not just a trading problem: every move pays gas, and a transaction submitted for block n cannot see the swaps that will occur in that same block.

We study this repositioning-frequency question in two parts. First, a continuous-time model shows how recentering rate trades off a structural

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concavity loss against an explicit per-move cost, and yields a closed-form optimal trigger within a simple one-parameter family of policies. Second, we build a no-anticipation block-time replay that evaluates strategies on historical swap order: decisions use only finalized history, fees accrue only when the post-swap price remains in range, and each recentering pays observed gas. Replaying 26 weeks of data from twelve Ethereum pools reveals three patterns. At retail scale, the full-window ranking is dominated by passive LP or HODL benchmarks; active rules do not win any pool at the reference setting, although they win 57 of 284 valid pool-weeks. At million-dollar scale in a major ETH/stable pool, where gas becomes much smaller relative to wealth, active rules still lose through inventory effects that mirror the theory's concavity drain in this window. Finally, an oracle allowed one block of foresight can change active returns by tens to hundreds of percentage points in high-fee pools, showing that backtests using the closing price of the current block can be misleading by a wide margin.

Keywords: Automated Market Maker, Concentrated Liquidity, Uniswap v3, Event-order Replay, No-anticipation, Gas, Decentralized Finance

1. Introduction

Decentralized finance (DeFi) runs financial services on public blockchains. Trading is a central use case: decentralized exchanges (DEXes) let users swap tokens through smart contracts, without routing orders through a centralized operator. A large share of DEX volume today flows through automated market makers (AMMs). In an AMM, there is no traditional order book. Instead, traders execute against a pool of two (or more) tokens held on chain; the exchange rate is determined by the pool's state according to a fixed mathematical rule. Liquidity providers (LPs) deposit assets into the pool beforehand and, in return, receive a pro-rata share of the trading fees paid by swappers.

Uniswap is the most widely used AMM family on Ethereum. Early versions, such as Uniswap v2, spread each LP's capital across the entire price curve: simple to operate, but capital-efficient only near the prevailing price. Uniswap v3 [1] changed the design by allowing *concentrated liquidity*. An LP chooses a finite price interval and supplies capital only inside it (Definition 3). Capital near the current price supports more trading volume per dollar deposited, but the position earns fees only while the market price remains inside the chosen interval. Once the price exits, fee income stops until the LP *recenters*—that is, moves the interval so that the current market price sits at its center again, typically by burning the old position and minting a new one on chain. In practice, maintaining a v3 position is like keeping a sliding price window aligned with a live market, and the question posed in the title is *how often* to recenter.

The naive answer—recenter whenever the price moves—keeps the window on the market and preserves fee opportunities, but it is costly on chain. Each recentering is a transaction: it consumes gas and may require swapping one token for the other to rebalance the position. It is also subject to block-time execution. A transaction included in block n cannot condition on the swaps that will be executed later in that same block, so a strategy that uses the block's closing price before the block is finalized is not implementable on Ethereum. Even if gas were free, repeatedly recentering into a moving price realizes a systematic loss from the nonlinear shape of the concentrated-liquidity payoff; our earlier conference work [2] showed that continuously chasing the price drains liquidity in a frictionless diffusion benchmark.

This paper turns that observation into a systems-style evaluation framework that a general systems or data researcher can inspect and reproduce. We pair a closed-form continuous-time benchmark with an event-order replay on historical chain data. The theory isolates *how recentering frequency trades off two costs*: a structural concavity loss that grows when the LP waits too long, and a transaction cost that grows when the LP moves too often. The replay then asks the same question on real swap streams: given block timing, gas prices, and transaction order, which policies actually pay off once fees, gas, and inventory effects are counted separately? The replay is deliberately not a full vault optimizer—it does not forecast fees or optimize dynamic range widths—but it makes the repositioning clock explicit, auditable, and measurable under a no-anticipation execution convention.

Our contributions are:

1. **A simple policy family with closed-form rates.** We index repositioning rules by a single *band* parameter: recenter once the price has moved a fixed fraction of the way to the range boundary, from continuous chasing to recentering only on exit. Under a standard diffusion price model, we derive a closed-form long-run decay rate for every band and drift (Theorem 3).
2. **A clean comparative static.** Without transaction costs, more fre-

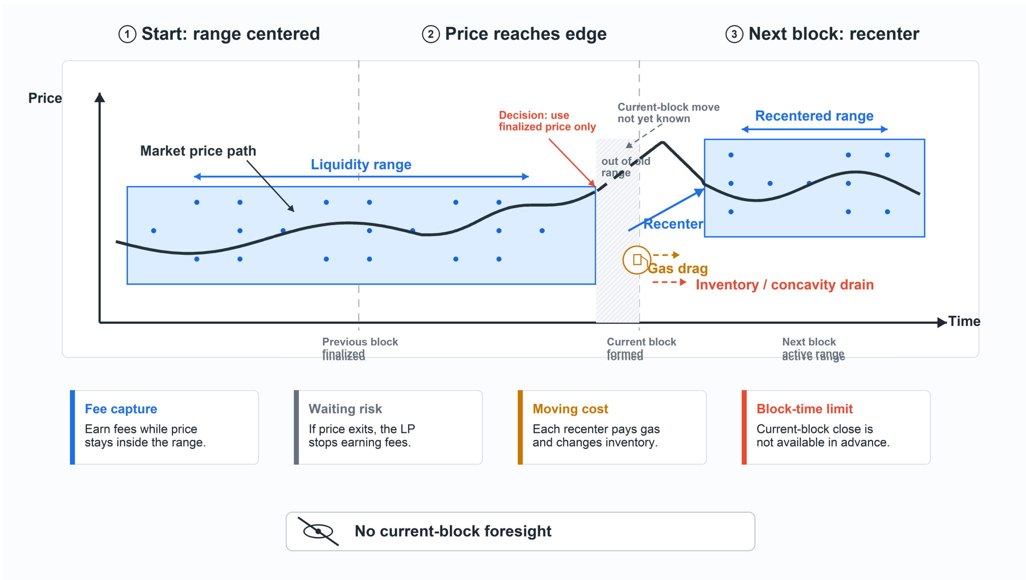

Figure 1: The recentering problem for a concentrated-liquidity LP. A narrow range can earn fees only while the realized price stays inside it. Recenter too often, and the LP pays gas and repeatedly changes inventory; wait too long, and the range can become inactive. Under block-time execution, the LP can use only finalized prices when deciding where to place the next range.

quent recentering always reduces the band-dependent concavity component (Theorem 4). Once a per-move cost is added, the optimum moves toward lazier triggers; for tight ranges it can collapse to "recenter only on exit" (Theorem 5).

3. **A no-anticipation replay model.** We define block-time policies that may use only finalized history before the current block, credit fees swap-by-swap when the realized post-swap price is in range, and charge observed gas per move (Section 8, Algorithm 1). Returns decompose into fee capture, gas drag, and inventory effects.
4. **A 26-week empirical study.** We replay twelve Ethereum Uniswap v3 pools (Mar–Aug 2024) across position sizes, range widths, and benchmark policies including passive holding, HODL, and a one-block oracle. Three results stand out under the stated replay conventions: (i) at retail scale, passive LP or HODL dominates the full-window ranking, while active rules win only in some weekly sub-windows; (ii) at million-dollar scale in WETH/USDC 0.05%, where gas is much smaller relative

to wealth, active rules still lose through inventory effects that mirror the theory's concavity drain in this window; (iii) allowing one block of foresight can change active returns by a first-order amount in high-fee pools, showing that backtests can overstate achievable performance when they use the current block's closing price.

The paper is organized as follows. Section 2 discusses related work. Section 3 states the problem. Sections 4–7 develop the continuous-time benchmark. Section 8 defines the replay model. Section 9 reports the chain-data evaluation. Section 11 summarizes the results.

2. Related work

AMMs and decentralized exchange. The formal study of automated market makers builds on earlier market scoring rules and on the constant-function market maker framework [3, 4]. Constant-product and constant-mean designs underlie Uniswap v2, Curve and Balancer [5, 6, 7], while Uniswap v3 introduces concentrated liquidity as a protocol design that lets LPs choose finite price intervals [1, 8, 9]. Angeris et al. [10] analyze Uniswap markets and their price-tracking properties, and Evans [11] studies LP returns in geometric mean markets. Empirical and equilibrium work studies how AMMs function as trading venues. Lehar and Parlour [12] analyze Uniswap as a decentralized exchange and compare AMM liquidity with limit-order-book markets. Aoyagi and Ito [13] study the coexistence of AMMs and limit order books and show how AMM liquidity interacts with asymmetric information and off-chain market structure. These papers explain why AMMs can support decentralized exchange and how their pool-level liquidity behaves. Our question is narrower: conditional on a concentrated liquidity position, how often should the LP move its range?

LP losses, LVR, and impermanent loss. The closest accounting concept is loss-versus-rebalancing. Milionis, Moallemi, Roughgarden and Zhang [14] define LVR as the shortfall of an AMM LP relative to a continuously rebalanced benchmark and derive closed-form running costs for AMMs, including concentrated liquidity. Subsequent work refines this benchmark and studies related rebalancing comparisons [15]. Cartea, Drissi and Monga [16, 17] study execution, predictable loss and optimal liquidity provision, incorporating costs associated with rebalancing and inventory adjustment. Loesch et al. [18] and Heimbach, Schertenleib and Wattenhofer [19] provide empirical evidence that Uniswap v3 LP performance is heterogeneous and that impermanent loss can exceed fee income. These works identify the economic losses borne by LPs. Our analysis isolates a different control margin: the loss induced by repeated recentering of a v3 range. This loss appears even before explicit gas is charged and is generated by the concavity of the concentrated-liquidity payoff each time the range is moved.

Strategic liquidity provision in Uniswap v3. Several papers formulate LP range choice as an optimization problem. Fan et al. [20] study strategic liquidity provision in Uniswap v3 and optimize dynamic allocation policies under price beliefs, risk preferences and reallocation costs. Fan et al. [21] analyze differential liquidity provision and use LP optimization to study how v3 contracts should partition price space. Online learning approaches treat liquidity placement as a repeated decision problem under observed pool states [22], while reinforcement-learning studies optimize active LP policies from simulated or historical market signals [23]. These papers consider richer strategy spaces and can be used to search for profitable policies. Our contribution is complementary. We restrict attention to a transparent one-dimensional family of band strategies, which makes the frequency of repositioning analytically visible. This yields closed-form comparative statics in the continuous-time model and a direct block-time implementation that can be replayed on historical transaction order.

Block-time execution and active liquidity. On-chain execution introduces timing frictions that are absent from continuous-time benchmarks. LPs do not observe the final within-block price before submitting a transaction for that block, and each reposition competes with other transactions for block space. The broader MEV literature studies how transaction ordering creates economic value and strategic behavior [24, 25, 26]. Just-in-time liquidity studies this block-level structure from a different angle: Wan and Adams [27] analyze LPs who mint and burn concentrated positions around individual swaps, and related work studies the strategic and empirical effects of such JIT provision [28, 29]. Our block-time model is not a JIT model and does not use mempool foresight. It imposes the opposite informational discipline: the strategy observes only finalized history and decides whether to recenter before the next block's realized swap path is known. This distinction is important because an oracle strategy can overstate fee capture and understate the cost of maintaining an active range.

Data-driven evaluation of DeFi strategies. Empirical DeFi studies increasingly evaluate mechanisms on realized on-chain event data rather than on simulated price paths. Loesch et al. [18] and Heimbach et al. [19] measure v3 LP outcomes from chain events; public event archives such as the Google BigQuery blockchain datasets, the XBlock Uniswap collection, Trading Strategy and Dune make swap-level replication practical [30, 31, 32, 33, 34]. Recent CLMM backtesting work builds a simulator for Uniswap v3 liquidity strategies and reports the sensitivity of portfolio values to strategy specification and tick-level state reconstruction [35]. Our replay differs in the axis it isolates: the action must be measurable with respect to finalized history, the oracle is explicitly marked as non-implementable, and every output decomposes fee capture, gas drag and inventory effects. This no-anticipation accounting interface is the link between the continuous-time band theory and the historical transaction-order evaluation.

3. The Problem

A typical workflow on a decentralized exchange involves three roles. A *trader* submits a swap against an automated market maker (AMM) and receives one token in exchange for another; the trade pays a fee that is routed to the pool. The *AMM* holds the pooled reserves and updates the exchange rate according to its trading rule whenever a swap is executed. A *liquidity provider* (LP) has deposited the two tokens into the pool beforehand and earns a share of the accumulated fees. The LP's objective is to deploy capital profitably: collect fees on active liquidity while limiting losses from adverse price moves and from the cost of adjusting the position. This paper concerns the LP's management problem in Uniswap v3, where capital is placed on a chosen price interval rather than across all prices.

In that setting, an LP selects a lower and upper price bound and supplies liquidity only inside the interval. While the market price remains inside that interval, the position is active and earns fees; once the price leaves the interval, the capital stays in the pool but no longer backs swaps, and fee income stops until the position is adjusted. Managing a concentrated position therefore reduces to a recurring decision: when, and how often, to move the interval so that it continues to cover the market price.

The most immediate response is to recenter the interval whenever the price moves. This rule keeps the position near the trading price and tends to preserve fee opportunities. The difficulty is that each reposition changes the

LP's token holdings. When the price of one asset rises, recentering typically shifts the portfolio toward the other asset—selling what has appreciated and buying what has depreciated. Repeating this adjustment after every small price move amounts to trading against the direction of the market. As shown in [2], this behavior can erode concentrated liquidity systematically even in a frictionless benchmark.

At the opposite extreme, the LP may hold the interval fixed and move it only after the price has exited. This *repositioning on exit* policy saves transaction costs and ignores fluctuations that reverse before the boundary is reached. Its cost is periods of inactivity: whenever the price lies outside the interval, no fees are earned. A narrow interval increases fee intensity near the current price but raises the frequency of exits; a wide interval reduces exits but deploys capital less efficiently. Neither extreme is dominant in general. The substantive question is the *frequency* of repositioning—whether to react to every move, to intermediate thresholds, or only to a full exit.

On a blockchain this question is further constrained by execution timing. Repositioning is submitted block by block, and the LP can act only on information already recorded on chain. The price realized in the current block is not known when the reposition decision for that block is made. A feasible strategy must therefore balance three measurable channels: maintaining fee capture by staying in range, limiting gas expenditure on moves that may not pay for themselves, and accepting the inventory and concavity losses created by price movement and recentering.

There is also a larger participation decision. A passive LP may hold the initial range and stop earning fees after the price exits; a HODL investor may hold the initial token mix outside the pool; or an operator may withdraw and not maintain the position at all. These alternatives are not band rules with a larger trigger; they are separate baselines evaluated alongside the active band family in Section 8. The continuous-time analysis below therefore answers a narrower diagnostic question: conditional on maintaining a symmetric concentrated-liquidity position through a band policy, how does the repositioning clock affect concavity loss and repositioning cost? The block-time replay then compares these active policies with passive LP, HODL and oracle benchmarks under realized swaps, gas and price paths.

The remainder of the paper formalizes this trade-off. Section 6 studies a family of repositioning rules in continuous time. Section 7 introduces a fixed cost per move. Section 8 states the block-by-block version suitable for evaluation on historical swap data.

4. AMM

AMMs allow Liquidity Providers (LPs) to provide liquidity passively to a trading pair X-Y of token X and token Y. One basic type of AMMs is the Constant Function Market Maker (CFMM) [4].

Definition 1 (CFMM)**.** A CFMM is specified by a continuously differentiable function $F : \mathbb{R}^2_{>0} \to \mathbb{R}$ with $\partial F/\partial x > 0$ and $\partial F/\partial y > 0$, called the invariant function/curve, its state (x, y) denoting the amount of deposited X and Y, and two state transition actions below.

CFMM quotes the price of X in Y by implicit differentiation along the invariant curve,

$$Z = -\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\partial F/\partial x}{\partial F/\partial y},$$

the infinitesimal exchange rate of X in Y, which is the definition of price. On a CFMM, the following two state transition functions are defined.

Definition 2 (State transitions)**.**

- Trade. User inputs $\Delta x > 0$ and AMM outputs $\Delta y > 0$ subject to the invariant curve constraint

  $$F(x + \Delta x, y - \Delta y) = F(x, y).$$

  Similarly for Δy.

- Add/withdraw liquidity. User inputs Δx and Δy ($\Delta x \Delta y > 0$, plus/minus sign for adding/withdrawing) subject to the constant price constraint

  $$Z(x + \Delta x, y + \Delta y) = Z(x, y)$$

Example 1 (Uniswap v2)**.** Uniswap v2 uses the invariant curve $F(x, y) = xy$. The price of X in Y in Uniswap v2 is then $Z = \frac{y}{x}$, which satisfies a basic property: The less the X, the pricier it gets. The constant price constraint translates to $\Delta y = Z\Delta x$, i.e. equal value of X and Y be provided/withdrawn.

There exists also Concentrated Liquidity Market Maker (CLMM) which permits LP to provide concentrated liquidity.

Definition 3 (Concentrated liquidity)**.** Let the current price of X be Z. A concentrated liquidity $l(Z, Z_1, Z_2)$ over price $[Z_1, Z_2]$ consists of

$$(x, y) = \begin{cases} \left(l\left(\sqrt{\frac{1}{Z_1}} - \sqrt{\frac{1}{Z_2}}\right), 0\right), & \text{if } Z < Z_1 \\ \left(l\left(\sqrt{\frac{1}{Z}} - \sqrt{\frac{1}{Z_2}}\right), l\left(\sqrt{Z} - \sqrt{Z_1}\right)\right), & \text{if } Z_1 \le Z \le Z_2 \\ \left(0, l(\sqrt{Z_2} - \sqrt{Z_1})\right), & \text{if } Z_2 < Z \end{cases} \tag{1}$$

amount of X and Y. In case of $Z < Z_1$ or $Z_2 < Z$, l is said to be out of range.

Equation 1 is well known and can be found for example in [14].

Two concentrated liquidities over the same range (Z_1, Z_2) are easy to combine. To combine concentrated liquidities over different ranges, we need the following decomposition theorem.

Theorem 1 (Liquidity decomposition)**.** *Let the current price be Z and $l(Z, Z_1, Z_2)$ be in range. Then for any interval (a, b), $Z_1 \le a \le Z \le b \le Z_2$, l can be decomposed as*

$$l(Z, Z_1, Z_2) = l(Z, Z_1, a) + l(Z, a, b) + l(Z, b, Z_2).$$

Proof. There are two things to prove: First, the amount of underlying tokens are the same. Second, the effect of the trade over the whole price range (Z_1, Z_2) is identical to the trade over the active liquidity range (a, b) as long as the post trade price $Z' \in (a, b)$.

The first part follows from

$$\begin{aligned} x(l(Z, Z_1, Z_2)) &= l\left(\sqrt{\frac{1}{Z}} - \sqrt{\frac{1}{Z_2}}\right) \\ &= 0 + l\left(\frac{1}{\sqrt{Z}} - \frac{1}{\sqrt{b}}\right) + l\left(\frac{1}{\sqrt{b}} - \frac{1}{\sqrt{Z_2}}\right) \\ &= x(l(Z, Z_1, a)) + x(l(Z, a, b)) + x(l(Z, b, Z_2)), \\ y(l(Z, Z_1, Z_2)) &= l\left(\sqrt{Z} - \sqrt{Z_1}\right) \\ &= l(\sqrt{a} - \sqrt{Z_1}) + l(\sqrt{Z} - \sqrt{a}) + 0 \\ &= y(l(Z, Z_1, a)) + y(l(Z, a, b)) + y(l(Z, b, Z_2)). \end{aligned}$$

For the second part, assume WLOG $Z < Z'$, i.e. we are buying X. If the trade uses $l(Z, Z_1, Z_2)$, by the invariant curve constraints before and after the trade, we have

$$\left(x(l(Z,Z_1,Z_2)) + \frac{l}{\sqrt{Z_2}}\right)\left(y(l(Z,Z_1,Z_2)) + l\sqrt{Z_1}\right) = l^2,$$
$$\left(x(l(Z,Z_1,Z_2)) + \frac{l}{\sqrt{Z_2}} - \Delta x\right)\left(y(l(Z,Z_1,Z_2)) + l\sqrt{Z_1} + \Delta y\right) = l^2,$$

which solves to

$$\Delta x = \left(1 - \sqrt{\frac{Z}{Z'}}\right)\left(x(l(Z,Z_1,Z_2)) + \frac{l}{\sqrt{Z_2}}\right) = \left(\sqrt{\frac{1}{Z}} - \sqrt{\frac{1}{Z'}}\right) l,$$
$$\Delta y = \left(\sqrt{\frac{Z'}{Z}} - 1\right)\left(y(l(Z,Z_1,Z_2)) + l\sqrt{Z_1}\right) = \left(\sqrt{Z'} - \sqrt{Z}\right) l.$$

If the trade uses $l(Z,a,b)$, we have

$$\Delta x' = \left(1 - \sqrt{\frac{Z}{Z'}}\right)\left(x(l(Z,a,b)) + \frac{l}{\sqrt{b}}\right) = \left(\sqrt{\frac{1}{Z}} - \sqrt{\frac{1}{Z'}}\right) l,$$
$$\Delta y' = \left(\sqrt{\frac{Z'}{Z}} - 1\right)\left(y(l(Z,a,b)) + l\sqrt{a}\right) = \left(\sqrt{Z'} - \sqrt{Z}\right) l.$$

This completes the proof of the second part. □

Theorem 1 implies that overlapping range liquidities can be added as follows. For $Z_1 < Z_3 < Z < Z_2 < Z_4$,

$$\begin{aligned} &l_1(Z,Z_1,Z_2) + l_2(Z,Z_3,Z_4) \\ =&l_1(Z,Z_1,Z_3) + (l_1+l_2)(Z,Z_3,Z_2) + l_2(Z,Z_2,Z_4), \end{aligned}$$

where $l_1 + l_2$ is the usual addition of functions.

5. Exogenous model and the chasing baseline

We use the following exogenous market model as a mechanism benchmark: the AMM price follows a geometric Brownian motion

$$dZ_t = \mu Z_t\,dt + \sigma Z_t\,dW_t, \tag{2}$$

with W_t a standard Brownian motion, and trades execute at the AMM price (zero slippage), so that when liquidity is rebalanced the exchange price coincides with Z_t. This idealization removes pool feedback, arbitrage latency and finite-depth execution. It is used to isolate the repositioning mechanism; the historical replay in Section 9 evaluates implementable policies directly on realized swap order.

The baseline strategy is *continuous chasing*: the LP continuously rebalances its range $[Z_t/\alpha, \alpha Z_t]$ so as to keep the AMM price at the center, holding a v3 position that is always freshly centered. The following deterministic decay was established in the conference version of this work.

Theorem 2 ([2]). *Under continuous chasing in the model of Equation 2, the liquidity process is deterministic and decays exponentially,*

$$L_t = L_0 \exp\left[-\frac{\sigma^2 t}{8(\sqrt{\alpha}-1)}\right].$$

The decay rate depends on the volatility σ only; the drift μ does not enter.

Proof. See [2], which states the driftless case; the proof there depends on the price path only through its quadratic variation, which is why μ does not appear. The decay rate $\sigma^2/(8(\sqrt{\alpha}-1))$ is recovered in Section 6 as the instantaneous concavity decay $\gamma(0)$ of marked-to-market wealth at the center of the range (Equation 5), and as the $b \to 0$ endpoint of the band family (Equation 6). □

Continuous chasing is the most active member of the repositioning family studied below. In the frictionless benchmark, its concavity component is the lower endpoint of the band-family rate (Theorem 4). This benchmark component excludes fee income, gas and block-time execution, which enter later through the replay accounting. The higher the volatility and the more concentrated the liquidity ($\alpha \to 1^+$), the faster this component accumulates.

6. A family of repositioning strategies

The chasing strategy of Theorem 2 re-centers the liquidity position on the AMM price at every instant, which requires unboundedly many repositions over any finite horizon. At the opposite extreme within the active-maintenance family, an LP may hold the position until the price leaves the range and then recenter. In this section, we embed these policies in a one-parameter family and compute, in closed form, the long-run marked-to-market rate associated with the concavity component of the benchmark, allowing the log-price to carry an arbitrary drift.

Throughout this section we work in the exogenous market model of Section 5: the AMM price Z_t follows the GBM $dZ_t = \mu Z_t\,dt + \sigma Z_t\,dW_t$, with $\log(Z_t/Z_0) = \nu t + \sigma W_t$ where $\nu := \mu - \sigma^2/2$. Let $h := \log\alpha$ be the half-width of a range in log-price and $A := 1/\sqrt{\alpha} = e^{-h/2}$.

Definition 4 (Band repositioning)**.** Fix a *band* $b \in (0, h]$. The LP holds a v3 position centered at the current AMM price; at every time the AMM price has moved by $\pm b$ in log from the center of the LP's current range, the LP repositions, re-centering on the prevailing price at no cost. Equivalently, with $\beta := b/h \in (0, 1]$, the LP repositions once the price has moved a fraction β of the way to the edge of its range. The limit $b \to 0^+$ ($\beta \to 0$) is continuous chasing; $b = h$ ($\beta = 1$) is repositioning only on range exit.

It is convenient to track the cycle in progress via the *internal log-price*

$$\xi_t := \log(Z_t/Z_c),$$

where Z_c denotes the center of the LP's current range; ξ_t measures how far the AMM price has drifted from where the LP last re-centered. By construction ξ takes value 0 at every reposition and stays within $[-h, h]$ during each cycle. Concretely, write $T_0 := 0$ and T_k for the k-th reposition time, $\tau_k := T_k - T_{k-1}$ for the duration of the k-th cycle, and $\xi_{T_k} \in \{-b, +b\}$ for the cycle's exit point. Since the LP's range center is reset at T_{k-1},

$$\xi_t = (\nu t + \sigma W_t) - (\nu T_{k-1} + \sigma W_{T_{k-1}}) \qquad \text{for } t \in [T_{k-1}, T_k],$$

so the strong Markov property of $\nu t + \sigma W_t$ at the stopping time T_{k-1} makes ξ, on each cycle, an independent copy of $\nu t + \sigma W_t$ started at 0 and run until first exit from $[-b, b]$. The pairs $\{(\tau_k, \xi_{T_k})\}_{k\geq 1}$ are therefore i.i.d.

6.1. Marked-to-market wealth and instantaneous decay

For a position centered at price Z_c with liquidity L, the value in token Y at internal log-price $\xi \in [-h, h]$ is, from Equation 1,

$$V(\xi) = \widetilde{L}\left(2u - Au^2 - A\right), \qquad u := e^{\xi/2} \in [A, 1/A], \quad \widetilde{L} := L\sqrt{Z_c}. \tag{3}$$

In particular $V(0) = 2\widetilde{L}(1-A)$, and a direct calculation gives the useful identity

$$\frac{V(b)}{V(-b)} = e^b, \tag{4}$$

which reflects the fact that the LP holds the relatively scarcer token at each boundary. Within a range (no rebalancing), Itô's lemma gives

$$d\log V(\xi_t) = (\log V)'(\xi_t)\,(\sigma\,dW_t + \nu\,dt) - \gamma(\xi_t)\,dt, \qquad \gamma(\xi) := -\frac{\sigma^2}{2}(\log V)''(\xi),$$

so $\gamma(\xi)$ is the instantaneous, concavity-driven decay rate of log-wealth from the diffusion alone. Under continuous chasing the position is always at $\xi = 0$, and a direct computation from Equation 3,

$$V(0) = 2\widetilde{L}(1-A), \quad V'(0) = \widetilde{L}(1-A), \quad V''(0) = \widetilde{L}\left(\tfrac{1}{2} - A\right),$$

$$(\log V)''(0) = \frac{V''(0)}{V(0)} - \left(\frac{V'(0)}{V(0)}\right)^2 = -\frac{A}{4(1-A)}, \qquad (\log V)'(0) = \tfrac{1}{2},$$

recovers the concavity rate of Theorem 2,

$$\gamma_{\text{chase}} := \gamma(0) = \frac{\sigma^2 A}{8(1-A)} = \frac{\sigma^2}{8(\sqrt{\alpha}-1)}. \tag{5}$$

6.2. The decay rate of the band strategy

Theorem 3. *Under band repositioning with band $b \in (0, h]$ and log-price drift ν, the marked-to-market benchmark rate is*

$$\bar{\gamma}(b;\nu) := \lim_{T\to\infty} -\frac{\log(V_T/V_0)}{T} \overset{a.s.}{=} \frac{R_{sym}(b)\,v(b;\nu)}{b^2} - \frac{\nu}{2}, \tag{6}$$

where

$$R_{sym}(b) := \log\frac{1-A}{1-A\cosh(b/2)}, \qquad v(b;\nu) := \frac{\nu b}{\tanh(\nu b/\sigma^2)}, \tag{7}$$

with the convention $v(b;0) := \lim_{\nu\to 0} v(b;\nu) = \sigma^2$.

Proof. Using the notation of the paragraph following Definition 4, set $R_k := -\log(V(\xi_{T_k})/V(0))$ for the log-wealth loss of cycle k. The pairs $\{(\tau_k, R_k)\}_{k\geq 1}$ are i.i.d., $\mathbb{E}[\tau_1] < \infty$ (computed below) and $\mathbb{E}|R_1| < \infty$ because $V(\pm b)$ are bounded away from 0 on $[-h, h]$. The renewal-reward theorem then gives

$\bar{\gamma}(b;\nu) = \mathbb{E}[R_1]/\mathbb{E}[\tau_1]$ almost surely; in the rest of the proof we suppress the cycle index and write τ, R, ξ_τ for cycle 1.

The cycle length τ is the first-passage time of $\nu t + \sigma W_t$ from 0 to $\pm b$. Writing $p_\pm$ for the probability of hitting $\pm b$ first, the standard formulas for drifted Brownian motion give

$$p_\pm = \frac{e^{\pm\nu b/\sigma^2}}{2\cosh(\nu b/\sigma^2)}, \qquad \mathbb{E}[\tau] = \frac{p_+ \cdot b - p_- \cdot b}{\nu} = \frac{b}{\nu}\tanh\frac{\nu b}{\sigma^2}, \tag{8}$$

with $\mathbb{E}[\tau] \to b^2/\sigma^2$ as $\nu \to 0$. Set $s \coloneqq e^{b/2}$ and $c \coloneqq s + s^{-1} = 2\cosh(b/2)$. From Equation 3,

$$\begin{aligned}\frac{V(b)\,V(-b)}{\widetilde{L}^2} &= \big(2s - A(s^2+1)\big)\big(2s^{-1} - A(s^{-2}+1)\big)\\ &= 4 - 4A(s+s^{-1}) + A^2(s^2+s^{-2}) + 2A^2 = (2-Ac)^2,\end{aligned}$$

where we used $s^2 + s^{-2} = c^2 - 2$. Since $Ac = 2A\cosh(b/2) \le 2A\cosh(h/2) = 1 + A^2 < 2$ for $b \le h$, $\sqrt{V(b)V(-b)} = 2\widetilde{L}\,(1 - A\cosh(b/2)) > 0$. Combined with Equation 4, this yields

$$\log V(b) = \log\sqrt{V(b)V(-b)} + \tfrac{b}{2}, \qquad \log V(-b) = \log\sqrt{V(b)V(-b)} - \tfrac{b}{2},$$

and hence

$$\mathbb{E}[R] = \log V(0) - \big(p_+ \log V(b) + p_- \log V(-b)\big) = R_{\text{sym}}(b) - \frac{b}{2}\tanh\frac{\nu b}{\sigma^2}.$$

Dividing by $\mathbb{E}[\tau]$ from Equation 8,

$$\bar{\gamma}(b;\nu) = \frac{\mathbb{E}[R]}{\mathbb{E}[\tau]} = \frac{\nu R_{\text{sym}}(b)}{b\tanh(\nu b/\sigma^2)} - \frac{\nu}{2} = \frac{R_{\text{sym}}(b)\,v(b;\nu)}{b^2} - \frac{\nu}{2}. \qquad \square$$

Remark 1 (Endpoints and limits)**.** Setting $\nu = 0$ in Equation 6 recovers the driftless rate

$$\bar{\gamma}(b;0) = \frac{\sigma^2}{b^2}\log\frac{1-A}{1-A\cosh(b/2)}.$$

As $b \to 0^+$, $R_{\text{sym}}(b) \sim Ab^2/[8(1-A)]$ and $v(b;\nu) \to \sigma^2$, so

$$\bar{\gamma}(0^+;\nu) = \gamma_{\text{chase}} - \frac{\nu}{2}.$$

The $-\nu/2$ term is the contribution of $\sqrt{Z_t}$-scaling: at the center the LP holds equal value in each token, so marked-to-market wealth in token Y absorbs half of the log-price drift; it is the same at every band and so plays no role in the choice of b (Theorem 4). At $b = h$, $1 - A\cosh(h/2) = \frac{1}{2}(1 - A^2)$, giving

$$\bar{\gamma}(h;\nu) = \frac{\nu}{h\tanh(\nu h/\sigma^2)} \log \frac{2\sqrt{\alpha}}{1+\sqrt{\alpha}} - \frac{\nu}{2}, \tag{9}$$

the decay rate of repositioning on exit.

6.3. The chasing point is the unique minimizer for every drift

We now show that $\bar{\gamma}(\,\cdot\,;\nu)$ is strictly increasing in the band for every ν. The driftless case ($\nu = 0$) admits a transparent Jensen argument from the integral representation, which we present first.

Lemma 1. *γ is even: $\gamma(-\xi) = \gamma(\xi)$ for all $\xi \in [-h, h]$.*

Proof. Writing $\gamma(\xi) = \frac{\sigma^2 A}{4} N(u)/d(u)^2$ with $N(u) := u^3 - 2Au^2 + u$, $d(u) := 2u - Au^2 - A$ and $u = e^{\xi/2}$, the reflection $\xi \mapsto -\xi$ sends $u \mapsto 1/u$. Direct substitution gives $N(1/u) = u^{-3}(1 - 2Au + u^2) = u^{-4}N(u)$ and $d(1/u) = u^{-2}d(u)$, so $\gamma(-\xi) = \frac{\sigma^2 A}{4} u^{-4}N(u)/u^{-4}d(u)^2 = \gamma(\xi)$. $\square$

Lemma 2. *γ has a unique critical point on $[-h, h]$, at $\xi = 0$, which is its strict global minimum; consequently $\gamma'(\xi) > 0$ for $\xi \in (0, h]$.*

Proof. With N, d as above, $d(u) > 0$ on $[A, 1/A]$ and $\gamma'(\xi) = 0$ is equivalent to $P(u) := N'(u)d(u) - 2N(u)d'(u) = 0$. A direct expansion gives the factorization

$$P(u) = (u-1)(u+1)\,Q(u), \qquad Q(u) := Au^2 + 2(1 - 2A^2)u + A. \tag{10}$$

The discriminant of Q is $4(1 - 4A^2)(1 - A^2)$. For $A \in [\frac{1}{2}, 1)$ it is non-positive, so Q has no real root; for $A \in (0, \frac{1}{2})$ the roots of Q have product $1 > 0$ and sum $-2(1 - 2A^2)/A < 0$, hence both are negative. In every case $Q(u) > 0$ for $u > 0$, so the only zero of P on $(0, \infty)$ is $u = 1$, i.e. $\xi = 0$. As $\gamma(\pm h)/\gamma(0) = 2/[A(1 + A)] > 1$, the unique critical point is the strict global minimum, and $\gamma' > 0$ on $(0, h]$. $\square$

Theorem 4. *For every $\nu \in \mathbb{R}$, $\bar{\gamma}(\,\cdot\,;\nu)$ is strictly increasing on $(0, h]$. Hence over the family of Definition 4, continuous chasing ($b \to 0^+$) uniquely minimizes this benchmark component at $\gamma_{chase} - \nu/2$, and repositioning on exit ($b = h$) maximizes it.*

Proof. Write $\bar{\gamma}(b;\nu) + \nu/2 = R_{\mathrm{sym}}(b)\, v(b;\nu)/b^2$ from Equation 6.

Case $\nu = 0$. Then $v(b;0)/b^2 = \sigma^2/b^2$, and the Green's function of σW on $[-b, b]$ started at 0 is $(b - |\xi|)/\sigma^2$, so

$$\bar{\gamma}(b;0) = \frac{\mathbb{E}\int_0^\tau \gamma(\xi_s)\, ds}{\mathbb{E}[\tau]} = \frac{1}{b^2}\int_{-b}^{b} \gamma(\xi)\,(b - |\xi|)\, d\xi = \int_{-1}^{1} \gamma(bt)\,(1 - |t|)\, dt. \quad (11)$$

Differentiating under the integral and using Lemma 1 (γ' is odd) to combine the contributions from t and $-t$,

$$\bar{\gamma}'(b;0) = 2\int_0^1 \gamma'(bt)\, t\,(1-t)\, dt,$$

and the integrand is strictly positive for $b \in (0, h]$ by Lemma 2, so $\bar{\gamma}'(b;0) > 0$. Hence the function $b \mapsto R_{\mathrm{sym}}(b)/b^2$ is strictly increasing on $(0, h]$.

Case $\nu \neq 0$. Define $f(x) \coloneqq x\coth(x)$. Then f is even with $f(0) = 1$, and for $x > 0$,

$$f'(x) = \frac{\sinh(2x)/2 - x}{\sinh^2(x)} > 0$$

since $\sinh(y) > y$ for $y > 0$. Hence f is positive and strictly increasing in $|x|$, and so is $v(b;\nu) = \sigma^2 f(\nu b/\sigma^2)$ in b (for $b > 0$). Since $R_{\mathrm{sym}}(b)/b^2$ is also positive and strictly increasing on $(0, h]$ (the driftless case above), the product $R_{\mathrm{sym}}(b)\, v(b;\nu)/b^2$ is strictly increasing in b, and so is $\bar{\gamma}(b;\nu)$. □

Remark 2 (Drift and numéraire effects)**.** The quantity $\bar{\gamma}(b;\nu)$ can be negative: with sufficient positive log-price drift, marked-to-market wealth measured in token Y grows because each unit of token X becomes worth more Y. Numerically, $\bar{\gamma}(0^+;\nu) < 0$ iff $\nu > 2\gamma_{\mathrm{chase}}$; the boundary $\bar{\gamma}(h;\nu) = 0$ defines a band-dependent threshold $\nu^*(h)$ at which drift exactly offsets the concavity component under repositioning on exit. The monotonicity of Theorem 4 is unaffected, but the sign of $\bar{\gamma}$ should not be read as a complete profitability statement because fees, gas and the decision to provide liquidity are outside this benchmark.

7. Repositioning under transaction costs

Theorem 4 says that, absent explicit costs, the concavity component within the band family is minimized by the most frequent recentering. Deployed strategies also face gas and possible slippage while rebalancing the two tokens. We now add a per-reposition cost and study the optimal trigger within the active band family. Passive holding or withdrawing from the position remains a separate baseline that must be compared through the full fee-and-cost accounting in Section 8.

Model each reposition as multiplying the position value by $1-\kappa$, where $\kappa \in (0,1)$ is the cost of one reposition as a fraction of position value. For gas-dominated costs $\kappa \approx c/V$ with c the fixed cost per transaction; for slippage-dominated costs κ is the effective spread paid to rebalance. With band b the reposition rate is $1/\mathbb{E}[\tau]$, so by the renewal-reward theorem the cost contributes a constant $\kappa' := -\log(1-\kappa)$ per cycle, and the net long-run decay rate is

$$\Gamma(b;\nu) := \bar{\gamma}(b;\nu) + \frac{\kappa'}{\mathbb{E}[\tau]} = \frac{v(b;\nu)}{b^2}\big[\kappa' + R_{\mathrm{sym}}(b)\big] - \frac{\nu}{2}, \tag{12}$$

the second equality using Theorem 3 and $1/\mathbb{E}[\tau] = v(b;\nu)/b^2$ from Equation 8.

As $b \to 0^+$, the bracket tends to $\kappa' > 0$ while $v(b;\nu)/b^2 \to \sigma^2/b^2 \to \infty$, so $\Gamma(b;\nu) \to +\infty$: continuous chasing is infinitely costly when every reposition is charged. The concavity term $\bar{\gamma}(b;\nu)$ increases in b (Theorem 4) while the cost term decreases, so Γ trades a lower repositioning clock against a larger benchmark concavity component.

Theorem 5. *Within the active band family, the cost-optimal band* $b^\star(\nu) := \arg\min_{b\in(0,h]} \Gamma(b;\nu)$ *is either an interior stationary point satisfying*

$$\frac{d}{db}\left\{\frac{v(b;\nu)\big[\kappa' + R_{sym}(b)\big]}{b^2}\right\} = 0, \tag{13}$$

or the boundary $b^\star = h$ *(repositioning on exit) when no lower point in* $(0,h)$ *improves the objective. The optimal band depends only on* α, κ *and the dimensionless drift-to-variance ratio*

$$\Pi := \frac{\nu h}{\sigma^2}, \tag{14}$$

in particular, it is invariant under simultaneous rescaling of ν *and* σ^2 *by the same factor; in the driftless case* $\Pi = 0$ *it is independent of* σ.

Proof. The function $\Gamma(\,\cdot\,;\nu)$ is continuous on $(0,h]$ with $\Gamma(0^+;\nu) = +\infty$ (since the bracket in Equation 12 is bounded below by $\kappa' > 0$ while $v(b;\nu)/b^2$ diverges), so a minimizer exists in $(0,h]$. The $-\nu/2$ term is b-independent and drops out of the FOC, leaving Equation 13.

Substitute $b = h\beta$ and $u = \nu b/\sigma^2 = \Pi\beta$ into Equation 12: $v(b;\nu)/b^2 = \sigma^2 f(\Pi\beta)/(h\beta)^2$ with $f(x) = x\coth x$, and $R_{\mathrm{sym}}(h\beta)$ depends only on (α,β). Hence

$$\Gamma(h\beta;\nu) + \frac{\nu}{2} = \frac{\sigma^2 f(\Pi\beta)\big[\kappa' + R_{\mathrm{sym}}(h\beta)\big]}{h^2\beta^2}.$$

The bracket and $f(\Pi\beta)/\beta^2$ depend on the LP's choices and parameters only through $(\alpha,\kappa,\Pi,\beta)$; the prefactor σ^2/h^2 and the constant $-\nu/2$ do not affect the minimizer. Therefore $\beta^\star(\Pi) = b^\star/h$ depends only on (α,κ,Π). □

The theorem asserts existence, not uniqueness, of the minimizer. In all numerical evaluations reported in this paper (Table 1 and the empirical calibration in Section 9) the objective has a single local minimum on $(0,h]$, so we refer to *the* optimal band without further qualification.

Mapping the replay to the theory. On chain, a fixed-dollar gas cost g per reposition corresponds to the relative cost $\kappa = g/W$ used in Theorem 5, with $\kappa' = -\log(1-\kappa) \approx g/W$ for small κ. Range concentration enters through $\alpha = e^h$ (equivalently $A = e^{-h/2} = 1/\sqrt{\alpha}$), where h is the log half-width as in Section 6. For ex-post comparison with the replay, we estimate a window-level drift-to-variance ratio from block-final ticks,

$$\hat\nu = \log\frac{P_T}{P_0}, \qquad \hat\sigma^2 = \sum_{n=1}^{N-1}\big[\log(P_n/P_{n-1})\big]^2, \qquad \hat\Pi = \frac{\hat\nu\, h}{\hat\sigma^2},$$

and substitute $(\kappa,\hat\Pi,h)$ into Equation 13. This calibration is *ex post*: it uses the full sample path and is not available to an online strategy. Its role is to test whether the lazy-trigger comparative statics of Theorem 5 appear in the replay when gas is measured rather than assumed.

For small bands, Equation 6 expands as

$$\bar\gamma(b;\nu) = \gamma_{\mathrm{chase}} - \frac{\nu}{2} + \left(\frac{\sigma^2\hat\gamma_2}{12} + \frac{\hat\gamma_{\mathrm{chase}}\,\nu^2}{3\sigma^2}\right) b^2 + O(b^4),$$

with $\hat\gamma_{\mathrm{chase}} \coloneqq A/[8(1-A)] = \gamma_{\mathrm{chase}}/\sigma^2$ and $\hat\gamma_2 \coloneqq A(1+2A)/[32(1-A)^2]$; the first b^2 term is the driftless concavity penalty and the second is the drift correction (note that drift enters through ν^2, so its effect is symmetric in sign). Combining with the b^{-2} cost term, Equation 13 yields the closed-form approximation

$$b^\star(\nu) \approx \left(\frac{12\,\kappa'}{\hat{\gamma}_2 + \dfrac{4\,\hat{\gamma}_{\text{chase}}\,\nu^2}{\sigma^4}} \right)^{1/4}, \tag{15}$$

valid when the right-hand side does not exceed h and the higher-order terms are small. Equation 15 is therefore used as a local comparative-static approximation: the optimal band widens as the fourth root of the cost, shrinks for tighter ranges, and shrinks further as drift magnitude grows (drift accelerates the cycle, so the cost-amortization benefit of waiting longer is partially offset). The approximation is accurate where it applies: at $\alpha = 2$ and $\kappa = 10\,\text{bp}$ it gives $\beta^\star \approx 0.54$ against the exact 0.53 of Table 1, and it degrades, as expected, near the boundary (at $\kappa = 100\,\text{bp}$ it gives 0.96 against the exact 0.91).

The two forces in Equation 12 move in opposite directions. The benchmark rate $\bar{\gamma}(b;\nu)$ rises with the band by Theorem 4, while the gas component $\kappa'/\mathbb{E}[\tau]$ falls as the LP waits longer between repositions. Table 1 reports the optimal normalized band $\beta^\star = b^\star/h$ obtained by solving Equation 13 exactly in the driftless case $\Pi = 0$, for a range of concentrations α and per-reposition costs κ. For the tightest range ($\alpha = 1.1$, a typical concentrated position), the constrained optimum within the maintained band family occurs at range exit for all but the smallest costs: the concavity penalty for waiting is small across such a narrow range, so earlier moves are hard to justify on this objective alone. As the range widens, the optimum moves into the interior—at $\alpha = 4$ and a 10 basis-point cost, the active band objective selects a trigger after the price has traversed about 40% of the way to the range edge. For nonzero drift, the optimal band depends on $|\Pi|$ alone (Theorem 5) and shrinks as $|\Pi|$ grows.

8. Block-time repositioning and evaluation

The previous sections treat the AMM price as a continuous-time process and allow the LP to reposition when a stopping time is reached. On-chain execution is coarser. Transactions are included block by block, the LP only observes prices that have already been finalized, and the price movement inside the block that contains the reposition is not known when the reposition

	cost per reposition κ			
α	1 bp	10 bp	100 bp	range half-width h
1.1	0.77	$1.00^{\dagger}$	$1.00^{\dagger}$	0.095
2	0.30	0.53	0.91	0.693
4	0.23	0.40	0.68	1.386
10	0.19	0.33	0.57	2.303
100	0.15	0.27	0.47	4.605

Table 1: Optimal normalized band $\beta^{\star} = b^{\star}/h$ at $\Pi = 0$ (driftless case) from the driftless first-order condition. A dagger † marks the boundary optimum $\beta^{\star} = 1$, i.e. repositioning on exit. For $\Pi \neq 0$ the optimum depends only on $(\alpha, \kappa, |\Pi|)$ and shrinks as $|\Pi|$ grows.

is submitted. We now define the discrete-time and event-order model used to evaluate implementable strategies.

8.1. No-anticipation block model

Let Δt be the block time and write $t_n = n\Delta t$. Denote by

$$P_n := Z_{t_n}$$

the AMM price at the end of block n. The information available before the LP acts in block n is

$$\mathcal{F}_{n-1} := \sigma(P_0, \dots, P_{n-1}),$$

or, in the transaction-order model, the sigma-field generated by all swap events in blocks $0, \dots, n-1$. The price change from P_{n-1} to P_n occurs after the reposition decision for block n has been made, and is therefore not available to the LP. This is the no-anticipation constraint.

Let C_{n-1} be the center of the LP's range before the decision in block n. The position is in range at a price P if

$$\left|\log \frac{P}{C_{n-1}}\right| \leq h, \qquad h := \log \alpha.$$

An admissible block strategy is a sequence of range centers $\{C_n\}_{n \geq 0}$ such that C_n is $\mathcal{F}_{n-1}$-measurable for every $n \geq 1$. Thus the center used during block n may depend on P_{n-1} and earlier prices, but not on P_n.

Definition 5 (Block band strategy)**.** Fix $b \in (0, h]$. The block band strategy with band b sets

$$C_n = \begin{cases} P_{n-1}, & \text{if } |\log(P_{n-1}/C_{n-1})| \geq b, \\ C_{n-1}, & \text{otherwise.} \end{cases}$$

The case $b = h$ is the block-time version of repositioning on exit. The formal limit $b \to 0$ corresponds to repositioning at every block, and is the implementable analogue of continuous chasing.

The definition deliberately uses P_{n-1} rather than P_n. An *oracle-exit* benchmark recenters using the final price of the block being entered: it observes the within-block outcome before acting, but still has no multi-block foresight. It is not admissible on chain and serves only to upper-bound any $\mathcal{F}_{n-1}$-adapted strategy. This distinction matters most when the block contains a large price move: a strategy may recenter to P_{n-1} and still fail to capture the whole block's fees if the realized price P_n leaves the new range.

8.2. Fee capture, gas cost, and marked-to-market wealth

We separate three effects in each block: fee income while the position is active, gas cost when the position is moved, and the change in the marked-to-market value of the inventory. Let V_n be the marked-to-market value of the LP position at the end of block n, measured in token Y in Equation 16 and in USD in the replay. In the data, USD valuation uses the block-implied ETH price from the WETH/USDC 0.05% pool together with the pool's token decimals. If the position used in block n is centered at C_n, define the block activity indicator

$$I_n(C_n) := \mathbf{1}\{|\log(P_n/C_n)| \leq h\}\,.$$

This is a coarse block-end convention. In the event-by-event replay used for data, the indicator is replaced by swap-level activity: a fee contribution is credited only for swaps whose realized post-swap price is inside the strategy's current range. This convention is intentionally adapted to finalized transaction order and avoids using the end-of-block price before it is known.

Let φ_n denote the dollar value of fees that the reference position would earn in block n if it were active throughout the block, and let g_n be the dollar gas cost of one reposition. Under the block-end convention, the realized net increment is

$$\Delta U_n = \varphi_n I_n(C_n) - g_n\,\mathbf{1}\{C_n \neq C_{n-1}\} + (V_n - V_{n-1}^{+}), \tag{16}$$

where V_{n-1}^{+} is the value immediately after any reposition at the beginning of block n. The first term is fee capture, the second is the fixed transaction cost, and the third is the mark-to-market inventory effect. Equation 16 is an accounting identity; different empirical specifications enter through the definitions of φ_n, g_n, and the event-level activity variable.

For comparison with Sections 6 and 7, it is convenient to record the per-block log-return criterion

$$J(\{C_n\}) := \liminf_{N\to\infty} \frac{1}{N\Delta t} \sum_{n=1}^{N} \left[\log \frac{V_n}{V_{n-1}^{+}} + \frac{\varphi_n I_n(C_n)}{V_{n-1}^{+}} - \frac{g_n}{V_{n-1}^{+}} \mathbf{1}\{C_n \neq C_{n-1}\} \right]. \tag{17}$$

We use J as an evaluation criterion on finite historical windows, not as a control problem to be solved: on data, the lim inf is replaced by the finite-window average. The first term is the concavity-driven component studied in Theorem 3; the last term is the discrete analogue of the cost contribution in Equation 12. The middle term is absent from the pure decay analysis and is what makes a concentrated position profitable in the first place.

8.3. Evaluation objective and benchmark policies

The criterion J of Equation 17 ranks admissible policies on a fixed event stream. The decision variable can be restricted to the band family of Definition 5, or enlarged to allow the LP to choose both the center and the width of the next range. In this paper the historical evaluation focuses on the restricted family and benchmark policies. On a finite window, the best band in the family is

$$b^{\star}_{\text{block}} \in \arg\max_{b\in(0,h]} J(\{C_n^b\}), \tag{18}$$

where C^b is generated by Definition 5; in practice we evaluate the finite-window average of J on a grid of bands.

Several benchmark policies are useful:

- *passive LP*: $C_n = C_0$ for all n; holds the initial range and never recenters after exit;
- *HODL*: hold the initial 50/50 token mix outside the pool; no fees and no gas, pure inventory exposure;
- *every-block recenter*: $C_n = P_{n-1}$ for every n;

- *fixed band*: Definition 5 with $b < h$;
- *recenter on exit*: Definition 5 with $b = h$;
- *periodic recenter*: $C_n = P_{n-1}$ every m blocks, otherwise hold;
- *oracle exit*: recenters using the final price of the block being entered; not $\mathcal{F}_{n-1}$-adapted.

Passive LP and HODL are baselines outside the maintained band family. The oracle prices one block of within-block foresight; it is not a mempool or JIT upper bound.

Algorithm 1 summarizes the event-order replay used in Section 9. Given an ordered swap stream and a strategy specification, it returns the fee, gas, inventory and net components of Equation 19.

8.4. Implementation mapping

The replay script implements Algorithm 1 as a deterministic accounting pass over each ordered pool stream. Each row in `block_time_results.csv` corresponds to one tuple of pool, range width, wealth scale and strategy. The output columns map directly to the accounting terms: `fees_usd` is the accumulated fee-capture term, `gas_usd` is the accumulated repositioning cost, `inventory_usd` is the residual marked-to-market inventory component, and `net_usd` is their sum relative to initial wealth. The corresponding `_bps` columns divide by initial wealth. The columns `repositions`, `active_share` and `bankrupt` report the operating behavior that produces the decomposition.

Remark 3 (Tight ranges and repositioning on exit)**.** The continuous-time cost model shows why the constrained active-band optimum can occur at range exit for sufficiently tight ranges: if h is small, the unconstrained cost-optimal band in Equation 15 can exceed the feasible interval $(0, h]$, and the boundary $b = h$ is chosen. The block model gives the same economic interpretation. A narrow range exits often, so gas is paid frequently; moving before exit raises the gas rate further, while the additional fee captured before exit may be small once block-time delay is respected. This does not imply that maintaining the position dominates passive holding; that comparison depends on realized fees, gas and inventory exposure.

Algorithm 1: Event-order replay with no-anticipation band policy

Input: Ordered swap stream $\{(B_k, \text{tick}_k, L_k, \ldots)\}_{k=1}^{K}$; initial wealth W_0; range half-width h; band b or benchmark label; gas schedule $g(B)$; ETH/USD lookup

Output: Fee, gas, inventory and net return over the window

Initialize center price $C \leftarrow P_1$ from the first swap; deploy liquidity from W_0;

for *each swap k in block order* **do**

  if *start of a new block* **then**

    Let $\text{tick}^{\text{prev}}$ be the final tick of the previous block;

    if *strategy is band policy* ***and*** $\left|\log(P_{k-1}/C)\right| \geq b$*, where* P_{k-1} *is the final price of the previous block* **then**

      Pay gas $g(B_k)$; if wealth insufficient, terminate and return bankruptcy;

      Recenter $C \leftarrow P_{k-1}$; redeploy liquidity at C;

    else if *strategy is every-block / periodic / oracle (per its rule)* **then**

      Apply the corresponding recentering rule (oracle uses the current block's final tick when the block closes);

  if *post-swap price P_k satisfies* $\left|\log(P_k/C)\right| \leq h$ **then**

    Credit fee share $L_{\text{own}}/(L_k + L_{\text{own}})$ times swap fee volume;

  Mark position to market at the post-swap price;

Aggregate fee, gas and inventory components; report net return;

8.5. Relation to LVR and block time

Loss-versus-rebalancing (LVR) measures the loss of an AMM LP relative to a continuously rebalanced reference portfolio [14]. The rate in Theorem 3 is not introduced through the same benchmark, but it has the same source: the value of a concentrated liquidity position is a concave function of price inside its range. Price variation therefore produces a systematic loss term. In the driftless chasing limit this term is

$$\gamma_{\text{chase}} = \frac{\sigma^2}{8(\sqrt{\alpha} - 1)},$$

which is proportional to the quadratic variation rate σ^2 and increases as the range becomes tighter.

Block time changes how this loss is realized. If a strategy recenters every block, then as $\Delta t \to 0$ the admissible block strategy approaches continuous chasing, but its gas rate explodes unless gas per reposition also vanishes. If a strategy recenters only on exit, the relevant clock is the first time the log-price crosses $\pm h$; the expected inter-reposition time is of order h^2/σ^2 in the driftless diffusion approximation. Thus the same volatility that increases the LVR-type loss also increases the frequency with which a narrow position must be moved.

The empirical block-time problem can therefore be read as a decomposition:

$$\begin{aligned} \text{net LP return} = \text{fee capture} - \text{gas drag} \\ - \text{LVR/concavity loss} + \text{drift and inventory effects.} \end{aligned} \tag{19}$$

Sections 6 and 7 provide the diffusion benchmark for interpreting the last three terms; Equation 16 and Algorithm 1 give the corresponding accounting identity and replay procedure for historical swap data.

9. Historical block-time replay

This section gives the empirical counterpart to the block-time formulation in Section 8. The purpose is not to calibrate a production LP vault, but to measure, on realized transaction order, the two cost channels that the theory separates: the explicit gas clock of Section 7 and the concavity drain of Theorem 4. The replay answers three systems questions. First, at what position scale, if any, does active range maintenance pay for itself once gas is charged at observed block prices? Second, how closely does the cost-optimal trigger implied by Theorem 5 from the *measured* cost ratio align with the best trigger in the replay? Third, how much of the loss of implementable strategies is due to the no-anticipation constraint itself, measured against an oracle with one-block foresight?

9.1. Data

We use Ethereum Uniswap v3 events from BigQuery over 26 weeks from 1 March 2024 to 30 August 2024. The raw event and transaction fields come from the Google Cloud public Ethereum and blockchain analytics datasets

Table 2: Dataset used in the 26-week block-time replay (1 March – 30 August 2024).

Pool	Type	Fee bps	Swaps	Mint/Burn
WETH/USDC 0.05%	volatile_major	5	1,215,421	38,586
WETH/USDC 0.30%	volatile_major	30	72,997	6,839
WBTC/WETH 0.05%	volatile_cross	5	246,857	9,650
WETH/USDT 0.05%	volatile_major	5	799,447	13,912
USDC/USDT 0.01%	stable_stable	1	199,990	5,094
DAI/USDC 0.01%	stable_stable	1	40,738	189
WETH/USDT 0.01%	volatile_major	1	828,572	8,364
WETH/USDC 0.01%	volatile_major	1	157,902	2,044
WETH/DAI 0.05%	volatile_major	5	145,417	832
WETH/USDT 0.30%	volatile_major	30	141,339	9,596
WBTC/WETH 0.01%	volatile_cross	1	79,032	1,272
WBTC/USDC 0.05%	volatile_cross	5	63,379	1,464

[30, 31, 36]. We screened alternative sources, including Trading Strategy's decentralized exchange data and the XBlock Uniswap v3 dataset, before settling on BigQuery because the replay requires block number, transaction index, log index, tick, $\sqrt{P}$ and pool liquidity in the same event stream [33, 32]. The sample contains twelve pools chosen to separate the main empirical dimensions: three WETH/USDC fee tiers, three WETH/USDT fee tiers, two WBTC/WETH fee tiers, one WBTC/stable pool, one WETH/DAI pool, and two stable/stable pools (Table 2). Swap events are ordered by block number, transaction index and log index, following Ethereum transaction ordering [37]. For each swap we observe the post-swap pool state: tick, $\sqrt{P}$, in-range liquidity and token amounts. Gas prices are joined at the transaction level. The ETH dollar price is implied block-by-block from the WETH/USDC 0.05% pool; external feeds such as Chainlink or Dune's curated price tables are natural substitutes for longer replays [38, 34].

9.2. Replay conventions

The accounting follows Equation 16 with the following conventions, stated explicitly because each one is a modelling choice rather than a measurement.

Small-position counterfactual. The reference LP does not change the realized price path. Its fee share for a swap with pool in-range liquidity L_{pool} is $L_{\text{own}}/(L_{\text{pool}} + L_{\text{own}})$, so the position dilutes itself but not the path. This is credible for retail-scale positions; we report scales up to \$1M in a tight range and treat larger positions as outside the model.

Fee crediting. A swap is credited all-or-nothing: the position earns its share of the swap's fee if and only if the realized post-swap price lies inside the strategy's current range. Swap volume is valued at the post-swap price.

Gas. Each reposition pays 300,000 gas units at the block-median gas price, converted to dollars at the implied ETH price. This covers the burn–mint–collect sequence of a recentering but excludes the swap needed to rebalance the two tokens; the replay therefore *understates* the cost of active strategies, which strengthens any conclusion that they are dominated.

Depletion. A strategy that cannot pay the gas cost of its next reposition is terminated: it stops trading and its final wealth is zero. Without this rule, decomposition components of failed strategies become accounting artifacts.

No-anticipation. At the start of block n, the strategy may inspect all finalized events through block $n-1$ and then fixes the range used while block n is replayed. It cannot condition on swaps later executed in block n or on the block's closing tick (Definition 5). The oracle benchmark below intentionally violates this convention by using the final tick of the block being entered.

For each pool we evaluate four symmetric half-widths $h = \log(1+r)$ for $r \in \{0.5\%, 1\%, 2\%, 5\%\}$. For a given h, we compare: *passive LP* (hold the initial range, never move), *HODL* (hold the initial 50/50 token mix outside the pool: no fees, no gas, pure inventory), *every-block* recentering, *periodic* recentering every 100 blocks, fixed-band rules with $\beta = b/h \in \{0.125, 0.25, 0.5, 0.75, 1\}$ where $\beta = 1$ is recenter-on-exit, and *oracle exit*, which recenters using the final price of the block being entered. The oracle is not implementable; its only role is to price the no-anticipation constraint. The position is marked to market event by event following Algorithm 1, and net return decomposes into fee capture, gas drag, and inventory effects exactly as in Equation 19.

9.3. Main patterns at the reference scale

Figure 2 summarizes the fixed-band family across all pools at the reference position size of \$100k. Two regularities are stable across pools. First, tight trigger bands are costly: at $\beta \leq 0.25$ the gas clock runs so fast that many configurations are terminated by the depletion rule before the window ends. Second, among active rules, wider bands are usually less fragile, but this does not make active maintenance the full-window winner. At the $\pm 1\%$ range and

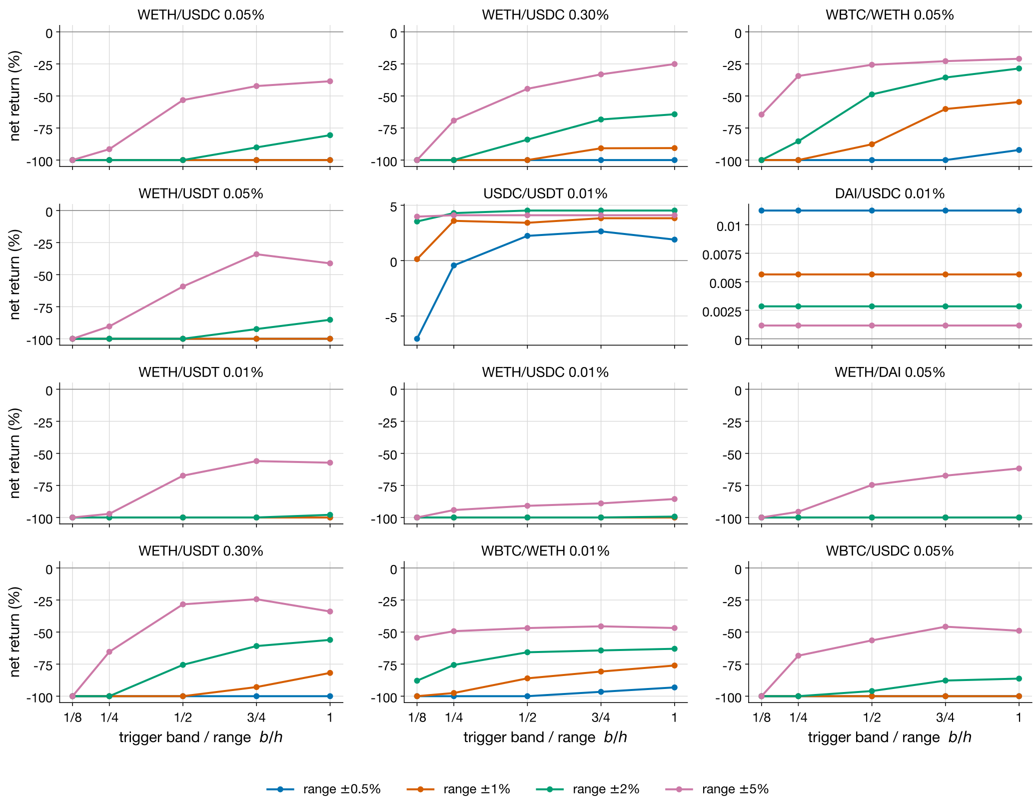

Figure 2: Net return of fixed-band strategies in twelve Uniswap v3 pools at the reference scale ($100k). Each panel reports net return, in percent of initial wealth, against the trigger band as a fraction of the range. Lines correspond to range half-widths. Stable/stable pools operate at a much smaller scale of returns.

$100k scale, passive LP is the best implementable strategy in ten pools and HODL is best in the remaining two; no active rule wins the full 26-week comparison.

Figure 3 decomposes the WETH/USDC 0.05% result at the $\pm 1\%$ range. Every-block recentering pays gas in essentially every block with a swap and is terminated by depletion (gas drag reaches 101% of initial wealth after 1,917 moves). The tight $\beta = 0.125$ band also terminates after 1,158 moves. The exit rule survives longer, makes 1,497 moves, and captures 86% of swaps, but it still terminates: fee capture reaches 147% of wealth, gas drag is 69%, and the inventory component is -178%. The realized inventory loss is the block-time counterpart of the concavity drain of Theorem 4 in this trending

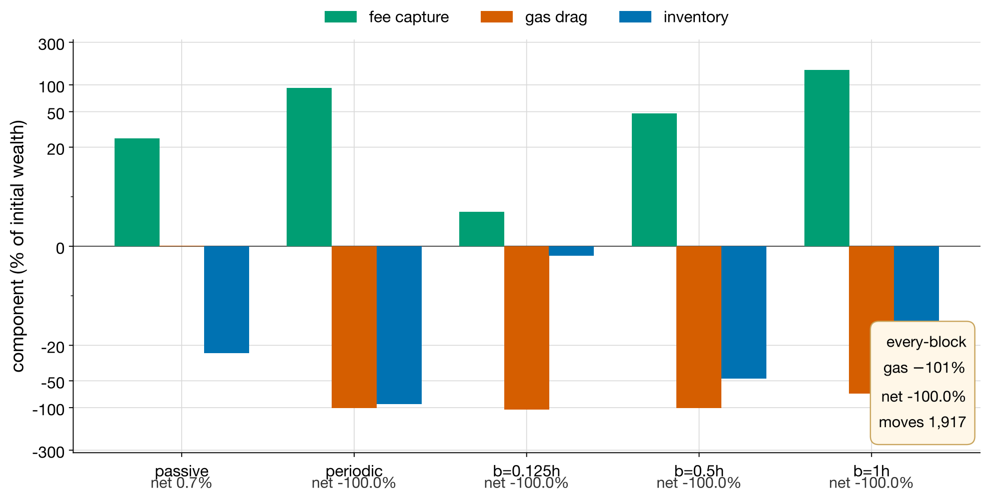

Figure 3: Fee, gas and inventory decomposition for WETH/USDC 0.05% with a ±1% range at the \$100k scale. Components are percent of initial wealth on a symmetric log scale. Strategies that cannot pay their next reposition are terminated at depletion, which caps gas drag near 100%. Every-block recentering is reported in the callout.

window. Passive holding earns 25% in fees, absorbs the inventory movement, and ends slightly positive.

The cross-pool view (Figure 4) and the operational view (Figure 5) complete the picture at this scale. The ranking of strategies is pool-dependent: passive LP wins most ETH/stable and stable/stable full-window comparisons, while HODL wins two low-fee volatile pools. High-fee pools still give intermediate bands a fighting chance in range-bound weeks (Table 6), even though active rules lose over the full window at this scale. Operationally, moving from $\beta = 0.125$ to $\beta = 1$ in WETH/USDC 0.05% raises the active-swap share from 2% to 86%. Reposition counts are not monotone after depletion because failed tight-band rules stop early; the more important measurement is that wider bands remain in range much more often before wealth is exhausted.

9.4. Position scale separates the two cost channels

All dollar-denominated quantities in the replay scale linearly with position size except gas, which is a fixed cost per transaction. The relative cost of one reposition is therefore $\kappa = g/W$ for position wealth W, and the theory of Section 7 predicts how the optimal trigger should respond. We exploit

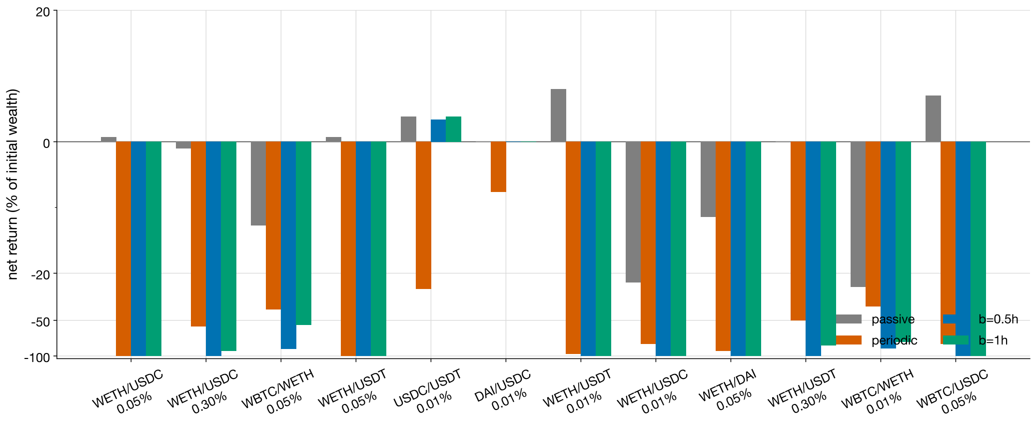

Figure 4: Net return by strategy and pool for the ±1% range at the $100k scale, in percent of initial wealth.

this to separate the two cost channels empirically: we rerun the replay at $10k, $100k and $1M, using the WETH/USDC 0.05% pool for the detailed decomposition and all twelve pools for the strategy-ranking check in Table 4.

Figure 6 shows the result for the exit rule at the ±1% range. As wealth grows from $10k to $1M, gas drag falls from depletion-level to 9.5% of wealth. The inventory component does not disappear: at $1M it is −270% of initial wealth, while fee capture is 187%. This is one central empirical finding in this sample: *the concavity and inventory channel identified in the frictionless theory remains visible at position scales where gas is no longer the binding cost*. Frequent recentering into a trending market repeatedly changes inventory exposure, so reducing transaction costs alone does not remove that loss in this window. Gas determines *where* active maintenance fails at small scale; the drain of Theorem 4 helps explain why it still fails at large scale in this pool and period.

Table 3 gives a diagnostic comparison with the theory. The median observed gas cost of one reposition is $8.14, so the measured cost ratio runs from $\kappa = 8.14$bp at $10k to 0.08bp at $1M. We form the ex-post inputs $(\kappa, \hat{\Pi}, h)$ as in Section 7 from block-final ticks in WETH/USDC 0.05% and substitute them into Theorem 5 to obtain the implied optimal trigger $\beta^\star$. At the measured $(\kappa, \hat{\Pi})$ this ex-post calculation gives the boundary optimum $\beta^\star = 1$ at tight ranges and remains near the boundary at ±5%. The replay is consistent with this lazy-trigger comparative static among active bands: at

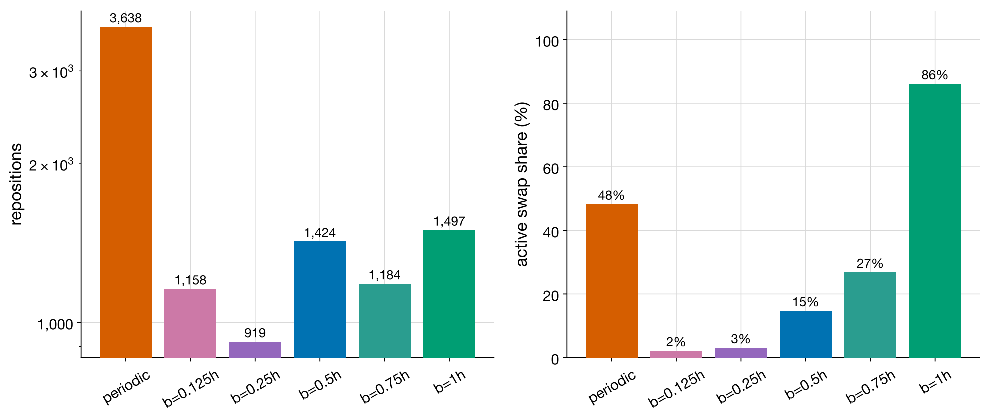

Figure 5: Reposition count (log scale, left) and active-swap share (right) for WETH/USDC 0.05% with a ±1% range. Wider trigger bands move less often but capture a much larger share of swap fees because the position stays in range longer.

Table 3: Measured cost ratio and the optimal trigger: theory versus replay (WETH/USDC 0.05%). κ is the median gas cost of one reposition divided by wealth; $\beta^\star$ theory solves the first-order condition at the measured $(\kappa, \hat{\Pi})$; $\beta^\star$ replay is the best fixed band in the replay ("–": all bands terminated by depletion).

Wealth ($)	Range	κ (bp)	$\beta^\star$ theory	$\beta^\star$ replay	Best band net %	Passive net %
100,000	±0.5%	0.81	1.00	–	-100.00	-1.69
100,000	±1%	0.81	1.00	–	-100.00	0.70
100,000	±2%	0.81	1.00	1	-80.59	1.49
100,000	±5%	0.81	0.98	1	-38.54	2.85
10,000	±1%	8.14	1.00	–	-100.00	1.26
1,000,000	±1%	0.08	1.00	1	-92.46	-3.53

±0.5% and ±1% every band is terminated by depletion, while at ±2% and ±5% the best surviving band is $\beta = 1$. However, the passive LP baseline still dominates the full-window active family.

Table 4 shows that the scale finding is not driven solely by the WETH/USDC 0.05% pool. At the ±1% range, exit-rule gas drag weakens as wealth grows in volatile pools, but the best implementable full-window strategy remains passive LP or HODL in almost all pool-scale cells. The only active best cell is the USDC/USDT 0.01% pool at $1M, where the return scale is small and the best band improves slightly over passive. This table does not prove a universal scale law; rather, it checks that the main ranking reported at $100k is stable across the three tested wealth scales within the same March–August

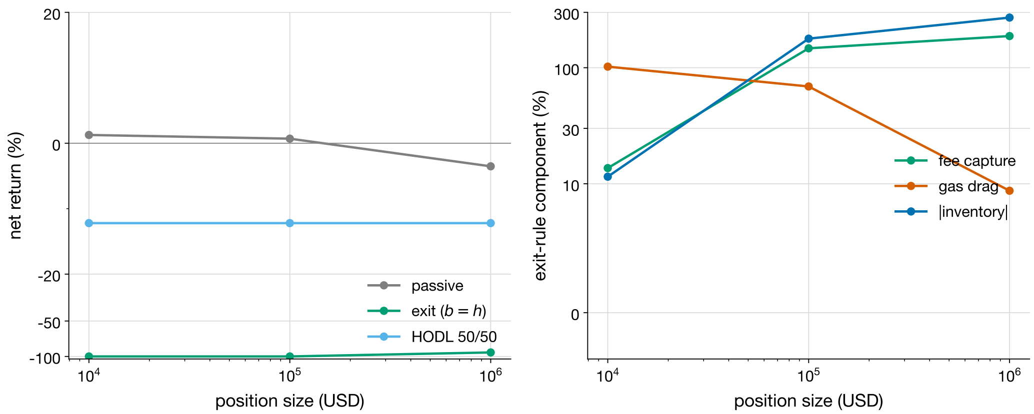

Figure 6: Position-scale sweep for WETH/USDC 0.05% at the ±1% range. Left: net return of passive, recenter-on-exit and HODL across position sizes. Right: decomposition of the exit rule; gas drag vanishes as the position grows, while the inventory (concavity) component persists.

2024 sample.

9.5. The price of no-anticipation

Table 5 compares, at the \$100k scale and ±1% range, the implementable benchmarks against the oracle-exit rule, which recenters using the final price of the block being entered. The oracle gap is large in high-fee volatile pools. In WETH/USDC 0.05% the oracle turns the exit rule's depletion into a +69.4% return; in the WETH/USDC 0.30% pool it earns +461.4% against the exit rule's −90.7%; in WETH/USDT 0.30% it earns +700.7% against −81.8%. In WBTC/WETH 0.05% the oracle improves the exit rule from −54.7% to −17.2%, but it does not reverse the sign. Stable/stable pools remain close to passive because little or no repositioning occurs. Two sample-level implications follow. First, in high-fee volatile pools a substantial part of the difficulty of active range maintenance is informational, not only frictional. Second, simulation studies that allow strategies to condition on the current block's closing price can overstate achievable LP returns by a first-order amount in these pools.

9.6. Robustness across weeks

The full window still covers a limited set of market regimes. To test whether the strategy ranking is an artifact of the full-window aggregation, we replay each pool week by week (±1% range, \$100k scale) and record the

Table 4: Per-pool scale check at the ±1% range. Exit columns report net return (% of initial wealth) for recenter-on-exit. Best columns report the best implementable strategy over the full window at each wealth scale.

Pool	10k exit	10k best	100k exit	100k best	1M exit	1M best
WETH/USDC 0.05%	-100.00	passive	-100.00	passive	-92.46	passive
WETH/USDC 0.30%	-100.00	passive	-90.66	passive	-80.57	passive
WBTC/WETH 0.05%	-100.00	passive	-54.71	passive	-48.89	hodl
WETH/USDT 0.05%	-100.00	passive	-100.00	passive	-96.06	passive
USDC/USDT 0.01%	13.49	passive	3.83	passive	1.22	$b = 0.25h$
DAI/USDC 0.01%	0.01	passive	0.01	passive	0.01	passive
WETH/USDT 0.01%	-100.00	passive	-100.00	passive	-100.00	hodl
WETH/USDC 0.01%	-100.00	hodl	-100.00	hodl	-100.00	hodl
WETH/DAI 0.05%	-100.00	passive	-100.00	passive	-100.00	hodl
WETH/USDT 0.30%	-100.00	passive	-81.85	passive	-77.45	passive
WBTC/WETH 0.01%	-100.00	hodl	-76.08	hodl	-74.80	hodl
WBTC/USDC 0.05%	-100.00	passive	-100.00	passive	-98.08	passive

best implementable strategy per valid pool-week. Weeks with fewer than 100 swaps are omitted, leaving 284 pool-weeks. Table 6 reports the counts. Passive holding wins 123 pool-weeks and HODL wins 104, so no active rule dominates within this sample. But the active family is not uniformly dominated either: fixed-band and periodic rules win 57 pool-weeks, concentrated in the WETH/USDT 0.30%, WETH/USDC 0.30%, and WBTC/WETH 0.05% pools. This is consistent with the comparative statics of Section 7: active maintenance becomes viable when fee intensity is high relative to the product of gas and realized trend.

9.7. Sensitivity to replay conventions

The replay conventions in Section 9.2 are intentionally explicit, but a reviewer still needs to know whether the main ranking is created by one convention. We therefore rerun the ±1%, $100k experiment under four local perturbations: halving gas to 150,000 units, doubling it to 600,000 units, replacing post-swap all-or-nothing fee crediting by a log-price path-overlap proxy, and replacing termination at an unaffordable reposition by skipping that reposition while keeping the old range live. Table 7 summarizes the result across the twelve pools. The magnitude of the exit rule changes materially, as expected: the median exit return ranges from -86.5% under 150k gas to depletion under 600k gas. The qualitative ranking is stable in this window: under the baseline, passive LP wins ten pools and HODL wins two;

Table 5: Net return (% of initial wealth) at the \$100k scale, $\pm 1\%$ range, over the full 26-week window. Oracle exit uses one block of foresight and is not implementable.

Pool	HODL	Passive LP	Exit ($b{=}h$)	Oracle exit	Every-block
WETH/USDC 0.05%	-12.21	0.70	-100.00	69.40	-100.00
WETH/USDC 0.30%	-12.31	-1.06	-90.66	461.35	-100.00
WBTC/WETH 0.05%	-13.75	-12.73	-54.71	-17.21	-100.00
WETH/USDT 0.05%	-12.15	0.71	-100.00	19.53	-100.00
USDC/USDT 0.01%	0.00	3.83	3.83	3.83	-100.00
DAI/USDC 0.01%	0.00	0.01	0.01	0.01	-100.00
WETH/USDT 0.01%	-9.75	8.05	-100.00	-80.96	-100.00
WETH/USDC 0.01%	-12.74	-23.96	-100.00	-71.21	-100.00
WETH/DAI 0.05%	-12.18	-11.45	-100.00	-88.54	-100.00
WETH/USDT 0.30%	-12.33	-0.05	-81.85	700.66	-100.00
WBTC/WETH 0.01%	-17.32	-26.15	-76.08	-30.25	-100.00
WBTC/USDC 0.05%	-1.62	7.02	-100.00	-71.34	-100.00

halving gas creates one active full-window win but leaves the broader passive/HODL ranking intact. Except in the doubled-gas case, where both exit and oracle rules often deplete, the median oracle gap remains positive, so the no-anticipation finding is not removed by these accounting perturbations.

9.8. Summary and limitations

Under the stated replay conventions and the sensitivity checks above, the sample supports four conclusions. (i) At retail scale in volatile major pools, gas is sufficient to make every active repositioning rule lose to passive LP holding over the full window; the cost-optimal response within the active family is the laziest trigger, consistent with the boundary calculation from Theorem 5 at the measured cost ratio. (ii) In WETH/USDC 0.05% at million-dollar scale, where gas is negligible, the loss of active rules persists and is attributable to the inventory/concavity channel—the block-time counterpart of the drain mechanism in Theorem 4 in this trending window. (iii) The no-anticipation constraint is expensive in high-fee volatile pools in this sample: one block of foresight can reverse the sign of the exit rule's return at identical cost. (iv) The ranking is regime- and pool-dependent: in high-fee pools during range-bound weeks, intermediate bands do win.

The main limitations are stated in Section 9.2 and in Section 10. In particular: the small-position counterfactual, approximate fee-crediting conventions, the fixed gas estimate that omits rebalancing swap costs, and a

Table 6: Best implementable strategy per valid pool-week ($\pm 1\%$ range, \$100k scale). Weeks with fewer than 100 swaps are omitted. "Other" are fixed-band and periodic active rules.

Pool	Weeks	Passive wins	HODL wins	Other
WETH/USDC 0.05%	26	13	11	2
WETH/USDC 0.30%	26	3	5	18
WBTC/WETH 0.05%	26	7	5	14
WETH/USDT 0.05%	26	13	12	1
USDC/USDT 0.01%	26	24	0	2
DAI/USDC 0.01%	26	26	0	0
WETH/USDT 0.01%	26	15	10	1
WETH/USDC 0.01%	17	2	15	0
WETH/DAI 0.05%	26	10	16	0
WETH/USDT 0.30%	26	2	5	19
WBTC/WETH 0.01%	7	2	5	0
WBTC/USDC 0.05%	26	6	20	0

single 26-week sample from one chain. The weekly sub-sample analysis addresses regime dependence within the window but not across market eras; replaying additional high-volatility periods and calibrating reposition gas from observed burn–mint–collect bundles in the mint/burn event data already collected are direct extensions of the same interface. Regeneration commands are listed in `REPRODUCIBILITY.md`.

10. Discussion

This paper studies repositioning policies for concentrated liquidity from two complementary viewpoints. The continuous-time model isolates the mechanism by which recentering frequency changes the concavity component of a v3 LP position. Embedding the price-chasing strategy of [2] in a band-indexed family, we obtained the closed-form benchmark rate

$$\bar{\gamma}(b;\nu) = \frac{R_{\text{sym}}(b)\, v(b;\nu)}{b^2} - \frac{\nu}{2},$$
$$R_{\text{sym}}(b) = \log \frac{1-A}{1-A\cosh(b/2)}, \qquad v(b;\nu) = \frac{\nu b}{\tanh(\nu b/\sigma^2)},$$

Table 7: Sensitivity of the full-window ranking to replay conventions at the $\pm 1\%$ range and \$100k scale. Active wins count implementable active policies: every-block, periodic, and fixed-band rules. Median exit is the median net return of recenter-on-exit across twelve pools; oracle gap is oracle-exit minus exit in percentage points.

Convention	Passive wins	HODL wins	Active wins	Median exit %	Median oracle gap (pp)
Baseline	10	2	0	-95.33	33.15
150k gas	9	2	1	-86.54	43.24
600k gas	10	2	0	-100.00	0.00
Path-overlap fees	10	2	0	-88.43	31.82
Skip unaffordable	10	2	0	-95.33	33.15

valid for arbitrary log-price drift $\nu = \mu - \sigma^2/2$ and any band from continuous chasing ($b \to 0$) to repositioning on range exit ($b = h$). The rate is strictly increasing in the band width for every drift, so absent explicit costs, continuous chasing minimizes this band-dependent concavity component. The driftless endpoint recovers the rate $\sigma^2/(8(\sqrt{\alpha} - 1))$ from [2]. This component is separate from the transaction cost of [17]: it is present even when repositioning is free, because the position value is concave in price.

Once a per-reposition cost is charged, the active-band objective changes. Very small bands become unattractive because the repositioning clock is too fast, while very large bands carry a higher benchmark concavity component. Within the maintained band family, the optimizer depends only on the concentration α, the relative cost κ, and the dimensionless drift-to-variance ratio $\nu h/\sigma^2$. In the driftless case it is independent of the volatility scale. For tight ranges, the constrained optimum can occur at the exit boundary; for wider ranges, an interior trigger can be preferable. The result is a mechanism benchmark for the repositioning clock; deployed LP profitability is determined by the block-time accounting with fees, gas and inventory exposure.

The systems contribution is the block-time replay formulation. In this model the LP observes finalized history through block $n - 1$ before acting in block n, fixes the range before the swaps in block n are replayed, earns fees only while the realized event-order price remains in range, and pays a fixed gas-unit cost priced at the observed block gas level. This separates implementable strategies from oracle benchmarks that use within-block information from the block being evaluated. It also gives the accounting decomposition

$$\begin{aligned}\text{net LP return} = {} & \text{fee capture} - \text{gas drag} \\ & - \text{LVR/concavity loss} + \text{drift and inventory effects},\end{aligned}$$

which is the interface between the closed-form mechanism and historical transaction-order simulation, implemented in Algorithm 1.

The 26-week replay turns this interface into three quantitative statements under the stated accounting conventions. At retail scale, active maintenance is dominated over the full window: at the reference $\pm 1\%$, \$100k setting, passive LP wins ten pools, HODL wins two, and no active rule wins. Stable/stable pools are different from volatile pools: fee and inventory magnitudes are much smaller, so active and passive LP returns can be close. In WETH/USDC 0.05% at million-dollar scale, where the measured cost ratio drops below one basis point, the active family still loses through the inventory channel, which is the realized counterpart of the concavity drain of Theorem 4 over this trending window. This suggests that the mechanism identified in the frictionless theory remains relevant after gas becomes small in that pool, but it is not a universal profitability claim. Finally, the oracle benchmark prices the no-anticipation constraint: one block of foresight turns the WETH/USDC 0.05% exit rule from depletion to +69.4%, and it produces much larger gains in the 0.30% fee tiers. Simulation studies that condition on the closing price of the block being evaluated can therefore overstate achievable LP returns by a first-order amount in high-fee pools.

Threats to validity. The replay relies on several explicit modelling choices: the small-position counterfactual, fee-crediting convention, fixed gas-unit assumption, USD valuation from the WETH/USDC pool, and a single 26-week sample from Ethereum mainnet. Section 9.7 checks that the full-window ranking is stable under local perturbations to gas units, fee crediting and depletion handling, but it does not calibrate the rebalancing swap cost from actual burn–mint–collect bundles. The scale ranking is reported across all twelve pools, while the detailed decomposition and ex-post calibration of $\hat{\Pi}$ remain centered on WETH/USDC 0.05%. The weekly analysis mitigates but does not remove dependence on the macro regime of March–August 2024. The continuous-time benchmark assumes exogenous GBM prices and a fixed symmetric range width. The band family conditions on maintaining an active concentrated-liquidity position; passive LP, HODL and withdrawal are separate baselines, and their ranking is regime-dependent. The role of the theory is to explain the repositioning clock and its cost channels; the replay determines whether an active policy is profitable after fees, gas and inventory exposure are all included.

Reproducibility. The replay is deterministic given the event data. The pool set, time window, BigQuery export templates, replay conventions and regeneration commands are documented in `REPRODUCIBILITY.md`. The swap, mint/burn and gas data are exported from the public Google BigQuery blockchain datasets [30, 31]; running `scripts/experiments/run_block_time_experiments.py` reproduces the LaTeX tables in Section 9, and `scripts/experiments/plot_figures.py` reproduces the figures.

11. Conclusion

This paper asks how to evaluate, on historical transaction order, how recentering frequency affects concentrated-liquidity LP returns under block-time constraints. We embed the conference chasing result in a one-parameter band family, derive a closed-form long-run decay rate under arbitrary log-price drift, prove that the band-dependent concavity loss is minimized by the most frequent recentering, and characterize the cost-optimal trigger once a per-reposition cost is charged. The optimizer depends only on range concentration, the relative cost ratio κ, and the dimensionless drift-to-variance ratio $\nu h/\sigma^2$.

We then define a no-anticipation block-time replay in which a strategy observes only finalized history before the current block, fixes its range before the current block's swaps are replayed, earns fees only when the realized event-order price remains in range, and pays observed gas per move. Replaying 26 weeks of swap order in twelve Ethereum pools shows that passive LP or HODL dominates the full-window reference setting, while active bands still win in some weekly sub-windows. In WETH/USDC 0.05% at million-dollar scale, where gas is much smaller relative to wealth, active rules still lose through the inventory channel—the realized counterpart of the concavity drain in Theorem 4 over this trending window. An oracle with one block of foresight can change active returns by tens to hundreds of percentage points in high-fee pools, so conditioning on within-block information materially overstates achievable LP performance in this sample.

At the evaluated full-window settings, passive holding beats every implementable active rule in our sample, yet weekly sub-samples show that intermediate bands can win in high-fee pools during range-bound weeks. The band-family benchmark and the replay decomposition separate the repositioning clock from deployed profitability: the theory helps identify which cost channel binds at which scale, and the data determine whether an active policy survives once fees, gas, and inventory exposure are all counted.

References

[1] H. Adams, N. Zinsmeister, M. Salem, R. Keefer, D. Robinson, Uniswap v3 core, Tech. rep., Uniswap (2021).
URL https://app.uniswap.org/whitepaper-v3.pdf

[2] R. Jiang, L. Wen, Y. Cao, Y. Ding, Chasing price drains liquidity, in: Mathematical Research for Blockchain Economy (MARBLE 2025), Lecture Notes in Operations Research, Springer, Cham, 2026, arXiv:2501.12583. doi:10.1007/978-3-032-13377-9_15.

[3] R. Hanson, Combinatorial information market design, Information Systems Frontiers 5 (1) (2003) 107–119. doi:10.1023/A:1022058209073.

[4] G. Angeris, T. Chitra, Improved price oracles: Constant function market makers, in: Proceedings of the 2nd ACM Conference on Advances in Financial Technologies, 2020, pp. 80–91. arXiv:2003.10001, doi:10.1145/3419614.3423251.

[5] H. Adams, N. Zinsmeister, D. Robinson, Uniswap v2 core, Tech. rep., Uniswap (2020).
URL https://uniswap.org/whitepaper.pdf

[6] M. Egorov, Stableswap: Efficient mechanism for stablecoin liquidity (2019).
URL https://curve.fi/files/stableswap-paper.pdf

[7] F. Martinelli, N. Mushegian, Balancer: A non-custodial portfolio manager, liquidity provider, and price sensor (2019).
URL https://balancer.fi/whitepaper.pdf

[8] Uniswap Labs, Uniswap protocol documentation, accessed 2026-06-11 (2024).
URL https://docs.uniswap.org/

[9] Uniswap Labs, Uniswap v3 sdk and pool state documentation, accessed 2026-06-11 (2024).
URL https://docs.uniswap.org/sdk/v3/overview

[10] G. Angeris, H.-T. Kao, R. Chiang, C. Noyes, T. Chitra, An analysis of uniswap markets, Cryptoeconomic Systems 1 (1) (2021). arXiv:1911.03380, doi:10.21428/58320208.c9738e64.

[11] A. Evans, Liquidity provider returns in geometric mean markets, Cryptoeconomic Systems 1 (2) (2021). arXiv:2006.08806, doi:10.21428/58320208.f1f9e4e7.

[12] A. Lehar, C. A. Parlour, Decentralized exchange: The uniswap automated market maker, Journal of Finance 80 (1) (2025) 321–374. doi:10.1111/jofi.13405.

[13] J. Aoyagi, Y. Ito, Coexisting exchange platforms: Limit order books and automated market makers, Journal of Political Economy Microeconomics 3 (3) (2025) 611–648. doi:10.1086/732831.

[14] J. Milionis, C. Moallemi, T. Roughgarden, A. Zhang, Automated market making and loss-versus-rebalancing (2024). arXiv:2208.06046, doi:10.48550/arXiv.2208.06046.

[15] C. McMenamin, J. Milionis, C. C. Moallemi, T. Roughgarden, Rebalancing-versus-rebalancing: Improving the fidelity of loss-versus-rebalancing (2024). arXiv:2410.23404.

[16] A. Cartea, F. Drissi, M. Monga, Decentralised finance and automated market making: Execution and speculation (2022). doi:10.2139/ssrn.4144743.

[17] A. Cartea, F. Drissi, M. Monga, Decentralized finance and automated market making: Predictable loss and optimal liquidity provision, SIAM Journal on Financial Mathematics 15 (3) (2024) 931–959. doi:10.1137/23M1602103.

[18] S. Loesch, N. Hindman, M. B. Richardson, N. Welch, Impermanent loss in uniswap v3 (2021). arXiv:2111.09192, doi:10.48550/arXiv.2111.09192.

[19] L. Heimbach, E. Schertenleib, R. Wattenhofer, Risks and returns of uniswap v3 liquidity providers, in: Proceedings of the 4th ACM Conference on Advances in Financial Technologies, 2022, pp. 89–101. arXiv:2205.08904, doi:10.1145/3558535.3559772.

[20] Z. Fan, F. Marmolejo-Cossío, D. Moroz, M. Neuder, R. Rao, D. C. Parkes, Strategic liquidity provision in uniswap v3, in: 5th Conference on Advances in Financial Technologies, Vol. 282 of Leibniz International Proceedings in Informatics, 2023, pp. 25:1–25:23. doi:10.4230/LIPIcs.AFT.2023.25.

[21] Z. Fan, F. J. Marmolejo-Cossío, B. Altschuler, H. Sun, X. Wang, D. C. Parkes, Differential liquidity provision in uniswap v3 and implications for contract design, in: Proceedings of the 3rd ACM International Conference on AI in Finance, 2022, pp. 9–17. arXiv:2204.00464, doi:10.1145/3533271.3561775.

[22] Y. Bar-On, Y. Mansour, Uniswap liquidity provision: An online learning approach (2023). arXiv:2302.00610.
URL https://arxiv.org/abs/2302.00610

[23] H. Zhang, X. Chen, L. F. Yang, Adaptive liquidity provision in uniswap v3 with deep reinforcement learning (2023). arXiv:2309.10129.
URL https://arxiv.org/abs/2309.10129

[24] P. Daian, S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, A. Juels, Flash boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability, in: 2020 IEEE Symposium on Security and Privacy, 2020, pp. 910–927. doi:10.1109/SP40000.2020.00040.

[25] K. Qin, L. Zhou, A. Gervais, Quantifying blockchain extractable value: How dark is the forest?, in: 2022 IEEE Symposium on Security and Privacy, 2022, pp. 198–214. arXiv:2101.05511, doi:10.1109/SP46214.2022.9833734.

[26] Flashbots, Flashbots documentation, accessed 2026-06-11 (2024).
URL https://docs.flashbots.net/

[27] X. Wan, A. Adams, Just-in-time liquidity on the uniswap protocol, sSRN working paper (2023). doi:10.2139/ssrn.4382303.

[28] K. Xiong, S. Wang, X. Liu, X. Cheng, Demystifying just-in-time liquidity attacks in uniswap v3, working paper (2023).
URL https://spiral.imperial.ac.uk/handle/10044/1/106835

[29] M. Bichuch, Z. Feinstein, G. Tsoukalas, The paradox of just-in-time liquidity in decentralized exchanges: More providers can lead to less liquidity (2023). arXiv:2311.18164.

[30] Google Cloud, Bigquery public dataset: Ethereum blockchain, accessed 2026-06-11 (2024).
URL https://console.cloud.google.com/marketplace/product/ethereum/crypto-ethereum-blockchain

[31] Google Cloud, Bigquery blockchain analytics datasets, accessed 2026-06-11 (2024).
URL https://cloud.google.com/blockchain-analytics/docs

[32] XBlock, Uniswap v3 dataset, accessed 2026-06-11 (2024).
URL https://xblock.pro/

[33] Trading Strategy, Uniswap trade dataset and decentralized exchange data, accessed 2026-06-11 (2024).
URL https://tradingstrategy.ai/

[34] Dune, Spellbook prices tables, accessed 2026-06-11 (2024).
URL https://docs.dune.com/data-catalog/curated/prices/overview

[35] A. Urusov, R. Berezovskiy, Y. Yanovich, Backtesting framework for concentrated liquidity market makers on uniswap v3 decentralized exchange (2024). arXiv:2410.09983.
URL https://arxiv.org/abs/2410.09983

[36] Google Cloud, Bigquery documentation, accessed 2026-06-11 (2024).
URL https://cloud.google.com/bigquery/docs

[37] G. Wood, Ethereum: A secure decentralised generalised transaction ledger, Tech. rep., Ethereum Foundation (2024).
URL https://ethereum.github.io/yellowpaper/paper.pdf

[38] Chainlink, Eth/usd price feed documentation, accessed 2026-06-11 (2024).
URL https://docs.chain.link/data-feeds/price-feeds/addresses